

Geospatial proposal assessment

USD describes a 3D model's shape and size. Geospatial information says where it belongs on Earth, so models, surveys and measurements can work together.

The question: can another implementer derive the behavior without guessing? The draft now defines the tested inputs and rules. Those choices still need group review.

Selected demonstrations work. A later audit found seven prototype issues. No additional model blocker was confirmed; this does not establish complete coverage.

A geographic location for a 3D model

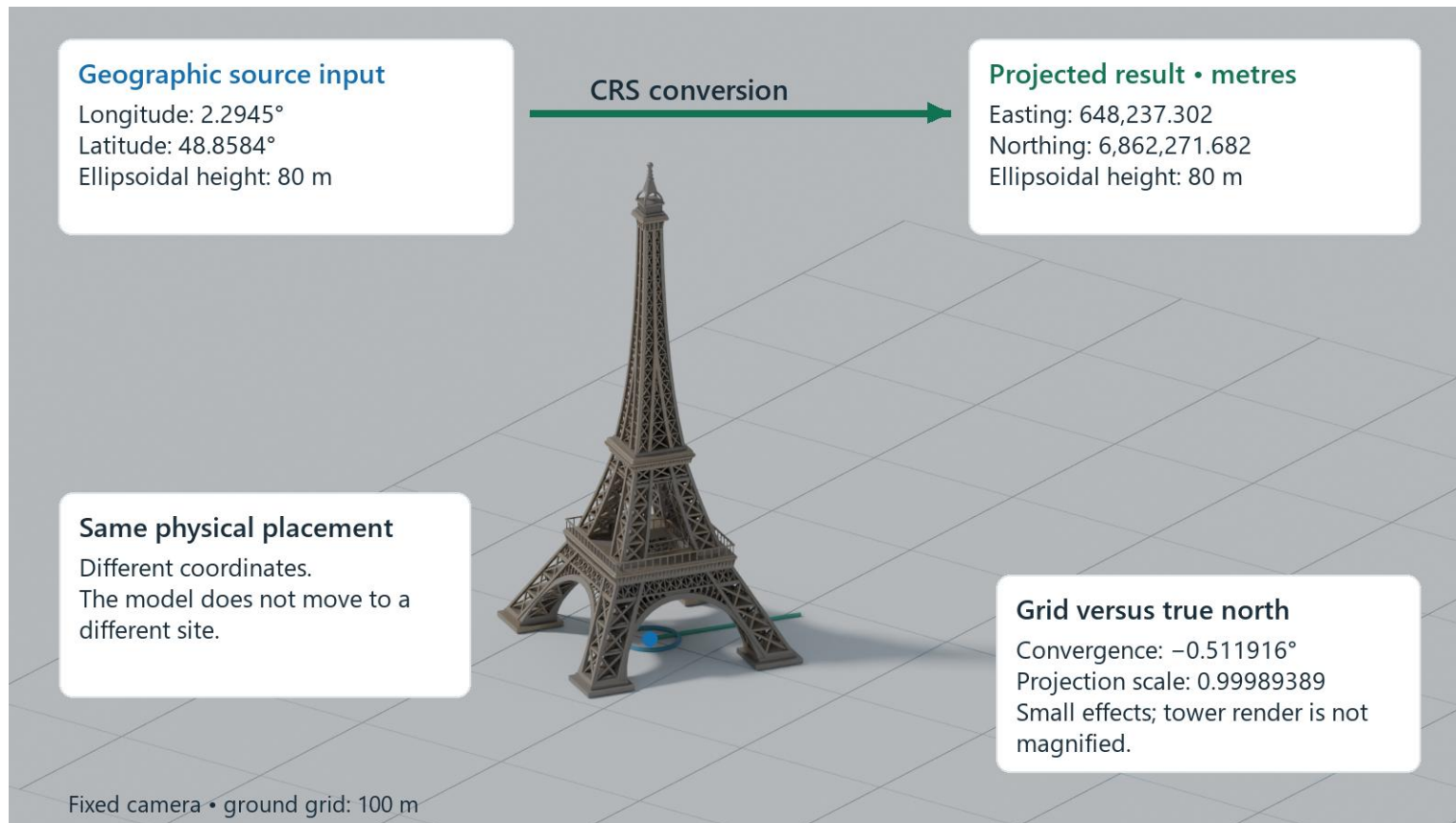
Can a reader distinguish the model's shape from its location on Earth?

Source information	Plain meaning
Model geometry	Shape and local distances, with declared model units and up direction. In USD: points, metersPerUnit and upAxis.
Coordinate reference	A CRS binding refers to WKT, the standard text description of the location's axes, units and height meaning.
Geographic origin	crs:position locates the model origin. The illustrative example uses longitude 2.2945 degrees, latitude 48.8584 degrees and ellipsoidal height 80 m.
Physical orientation	crs:orientation says which way the model faces and tilts relative to local east, north and up at its origin.

A reader has enough source information to place the model, without interpreting longitude or latitude as a distance or adding computed results to the source schema.

One physical place, different coordinates

Does changing coordinate systems preserve physical placement?



Observed

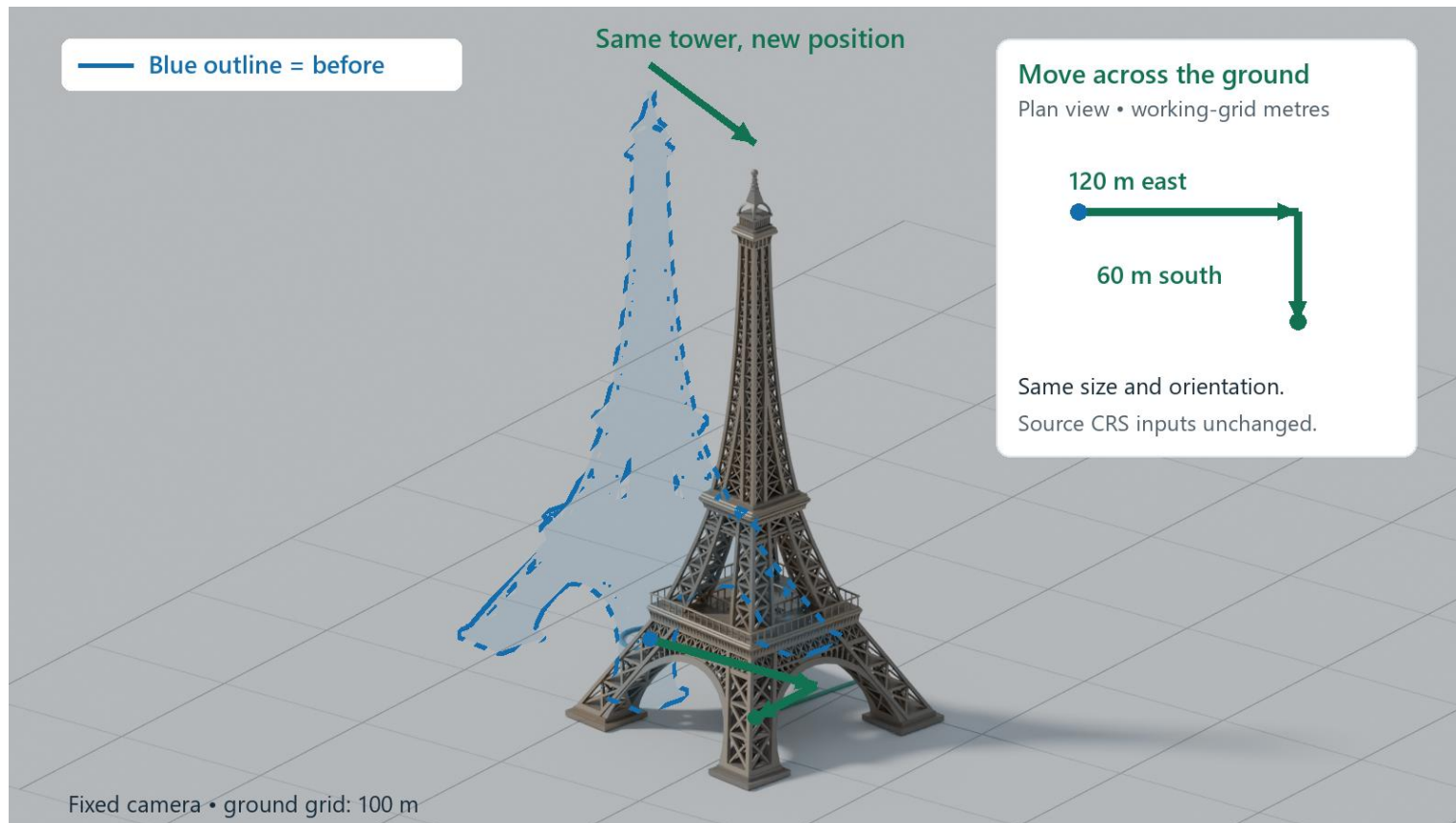
The illustration converts longitude, latitude and height into map coordinates. The physical site stays the same; map north and local scale can differ.

What it establishes

The proposed rules distinguish a new coordinate representation from a physical move. An image explains this distinction; numerical controls must test it.

Ordinary model adjustments after placement

In which coordinates do an author's scale, rotation and translation operate?



Observed

The illustration applies scale 1.15, rotation 20 degrees, then moves 120 m east and 60 m south. A child-transform control separates two interpretations by 4.222 m.

What it establishes

Order changes the result. The control makes the proposed child-transform choice reviewable; matching readers do not establish agreement.

The same rules in two placement readers

Can separate implementations derive the same coordinates from the proposed inputs and rules?

France: CC49 to Lambert-93, metres

Resolved component	Python reader	C++ reader
Map east coordinate	661,101.109417	661,101.109417
Map north coordinate	6,857,834.256536	6,857,834.256536
Height	37.298000	37.298000

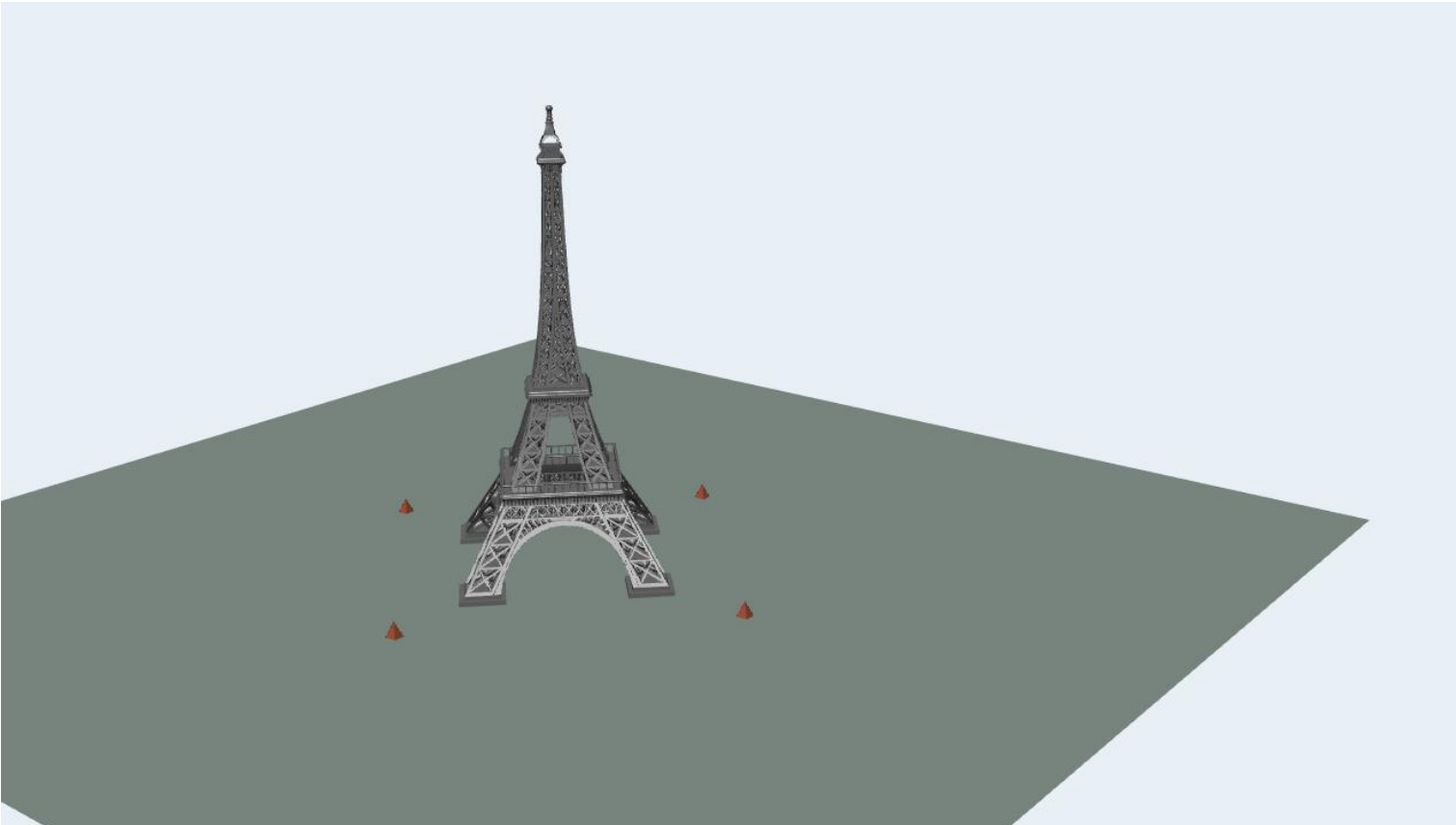
Colorado: State Plane to site grid, US survey feet

Resolved component	Python reader	C++ reader
Map east coordinate	3,108,871.450101	3,108,871.450101
Map north coordinate	1,206,317.469403	1,206,317.469403
Height	5,484.677949	5,484.677949

Both readers use PROJ, the same CRS conversion library, and share native dataset decoding.

Ordinary USD geometry for rendering

Can a renderer use the result without implementing geospatial evaluation?



Observed

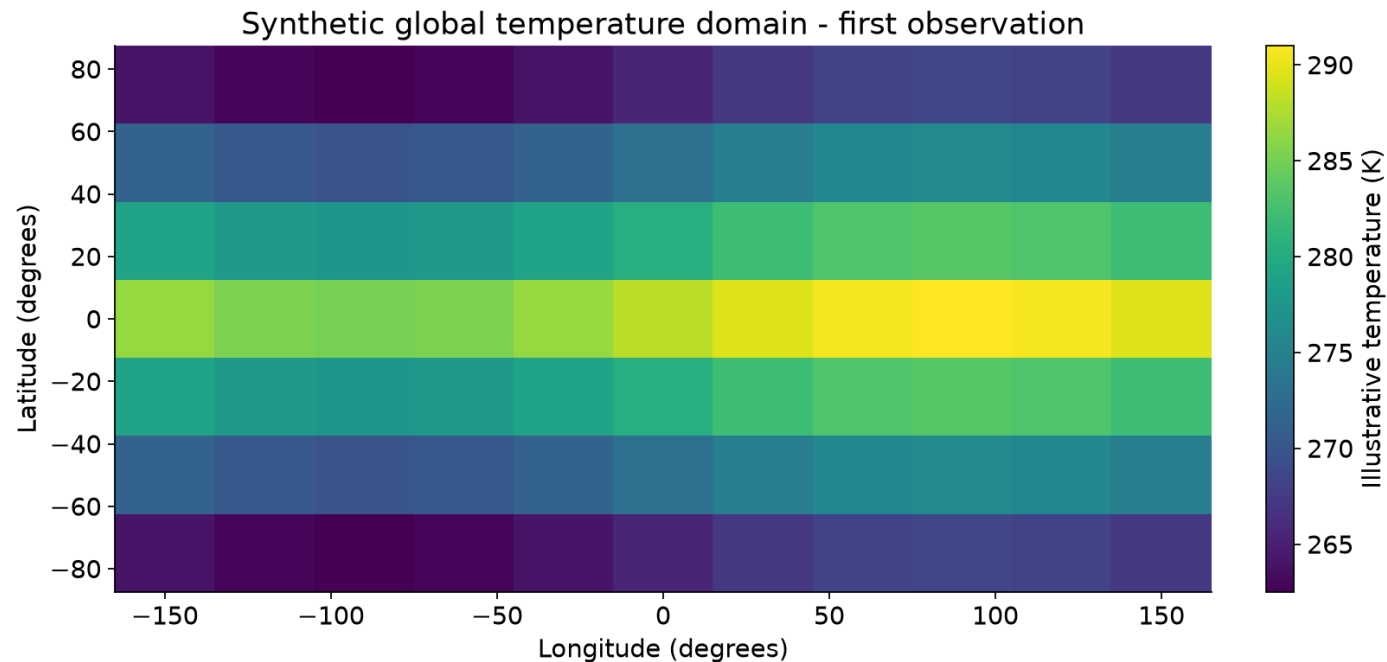
The C++ reader's resolved Eiffel copy reaches Hydra/Storm, a USD rendering path. This image is actual output from the recorded run.

What it establishes

This case shows compatibility with an ordinary USD renderer. A separate nested-geometry failure prevents a claim that every bake is correct.

Measurements stay paired with their locations

Can the scene make a native measurement dataset usable without rewriting its values or guessing what its coordinates mean?



The synthetic global temperature case returns both coordinate forms for 154 samples and preserves the native values and two observation times.

The heatmap contains synthetic temperatures in kelvin. A different array layout mispairs four of six values and locations. This passing case does not establish support for every layout.

What the later audit changes

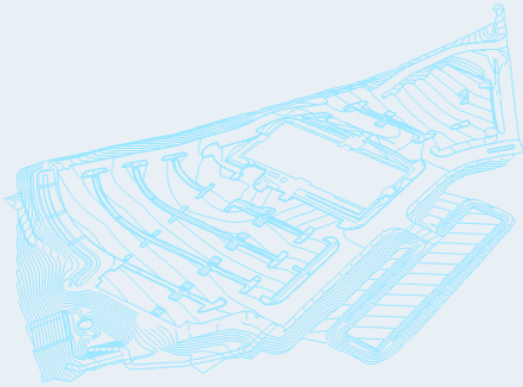
The recorded suite passed, then the audit exercised cases it had missed.

Affected behavior	Confirmed issue
Surface lighting directions	A private sampling shortcut has no justified error bound. One projected case differs by about 1.24 degrees.
Repeated models	A referenced model outside the bound subtree fails instead of receiving the instancer's placement.
Baked geometry	Two source triangles become three output triangles. The check misses the extra geometry.
Measurement locations	A different array dimension order pairs four of six values with the wrong locations. Both readers share this decoder.
Other scene information	The bake removes an unrelated MotionAPI schema application, changing other scene meaning.
Error reporting	Reports omit required per-component errors and separate angular errors.
Invalid input detection	An invalid placement attribute on an inherited-only child is silently ignored.

Repair these against the existing proposed rules. If repair reveals truly unspecified meaning, flag that separately.

Appendix: A construction site's own survey grid

Can a project's local survey grid participate as a defined CRS?



Observed

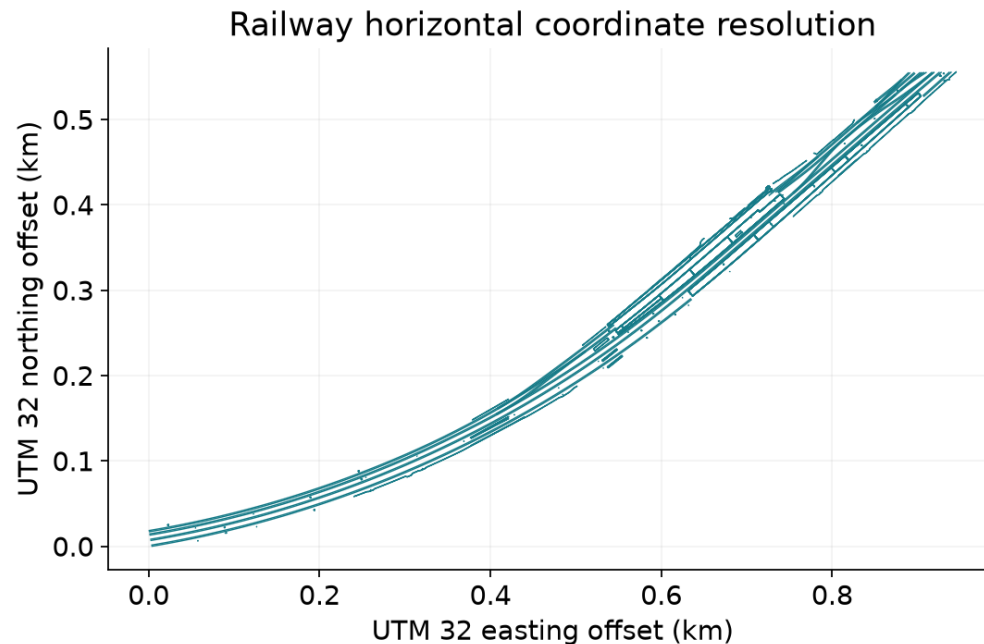
The original LandXML survey breaklines resolve and reach an ordinary USD bake rendered by Storm.

What it establishes

The case supports a site's calibrated CRS using partner survey lines. It does not establish independent survey accuracy.

Appendix: Useful horizontal placement without invented height

Can incomplete source metadata still support an honestly limited workflow?



Observed

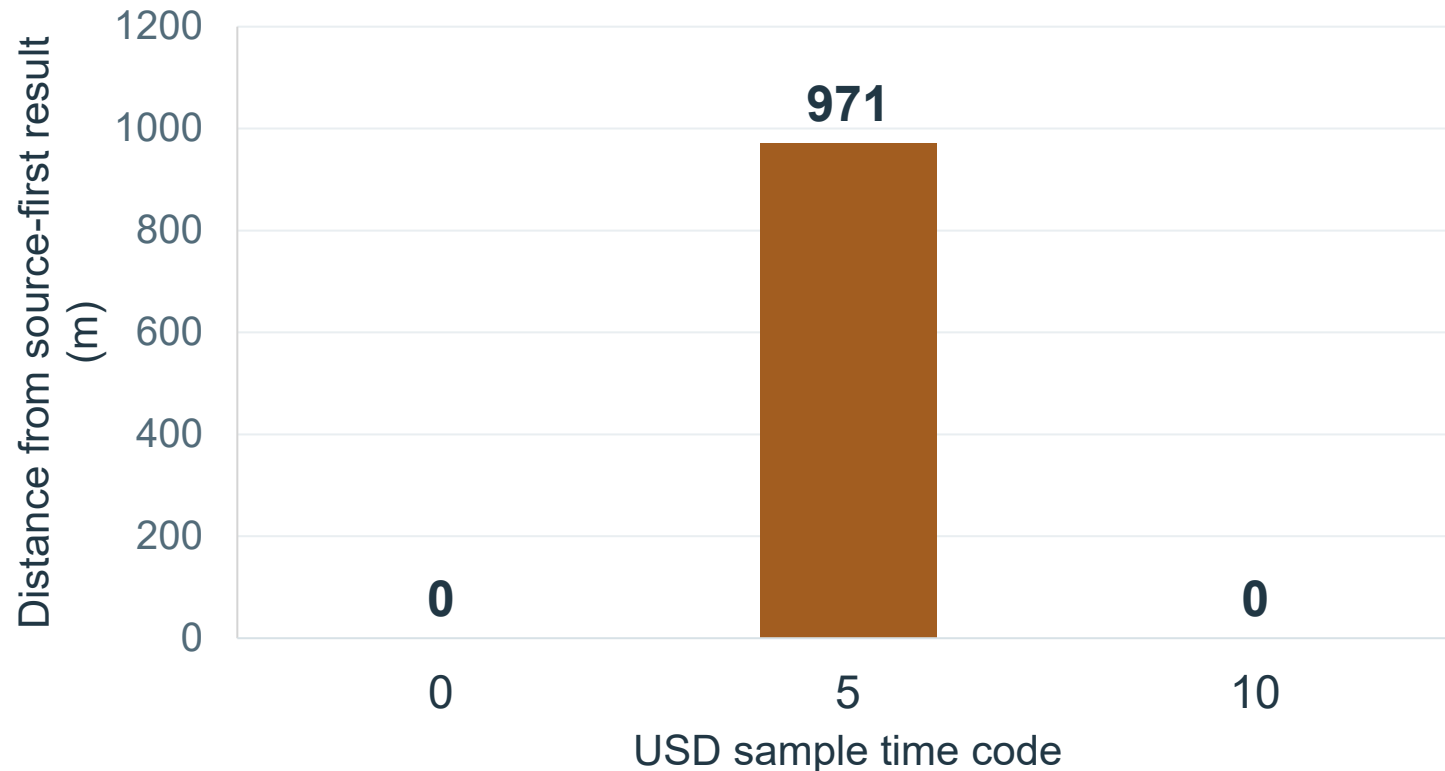
The horizontal railway copy resolves 15,822 vertices across 2,386 parts. The plot uses display-only offsets.

What it establishes

This case exercises visible limits on interpretation, not a silent assumption that every three-number tuple is a valid geographic 3D position.

Appendix: Animation evaluation order

**Interpolate the authored source location first, then convert coordinates.
Interpolating already-converted endpoints can produce a different path.**



Why the order matters

The distinguishing control makes the ordering rule testable. Preserving sample times alone cannot prove preservation of the original trajectory between samples.

The bake preserves time codes 0, 5, 10 and 48 time codes per second. It provides no original-trajectory guarantee between those samples.

Appendix: Placement choices under review

Topic	Proposed meaning to review
Physical orientation	Record model orientation relative to local east, north and up. A different map projection does not redefine it.
The object with its own georeference	Its ordinary scale, rotation and translation adjust placed geometry in a fixed project frame.
Child objects	Child transforms define model-local geometry. Their offsets follow model axes instead of being silently reinterpreted as map axes.
Independently georeferenced repeated models	Keep a prototype's own geographic placement and apply each instance effect once.
Geographic queries and scene geometry	Return longitude/latitude/height for geographic queries. Cartesian geometry and bounds use the datum's Earth-centred frame.

These are definitions in the candidate, not new inferred behavior supplied by the demonstrations.

Appendix: Data and declaration choices

Topic	Proposed meaning to review
Native data association	Name the asset, format, field and coordinate domain. Preserve the association between values, locations and observation times.
WKT string normalization	A prescribed text form supports token comparison. Different normalized texts can still describe equivalent CRSs.
Semantic CRS comparison	Specify equivalence criteria and the comparison domain, accounting for relevant axes, units and metadata.
Dependency declaration using Profiles	Declare the composed scene's need for geospatial interpretation, including unloaded content. Writers and assemblers maintain the declaration.
Baked output context	Keep a Cartesian CRS context without an active placement binding, so readers do not apply georeferencing a second time.

Coordinate epochs and scene-authored operation/resource controls remain deferred without foreclosing later support.

Appendix: Evidence and its limits

Path or check	Recorded evidence
Python and C++ readers	213,231 coordinate results per reader. Maximum measured difference 3.39e-08 m. They share PROJ and native dataset decoding.
Omniverse stage geometry integration	Reuses Python placement. Recorded readback covers 177,864 geometry vertices and 36 edit, failure and recovery checks, across 35 jobs.
Hydra / Storm	Two selected ordinary USD bakes reach native rendering. This consumer path uses the C++ placement results.
Recorded tests	64 regressions and 61 distinguishing controls passed. The later audit exposes cases the suite missed.
Source preservation	Original inputs and 40 executed source-file hashes remain unchanged. The immutable receipt identifies the executed content.
Bake readbacks	11 selected exports were checked. The later nested-geometry case demonstrates that the original inventory check was incomplete.

There are two placement readers. Omniverse reuses Python with a verified stage geometry sink. Hydra consumes a C++-resolved ordinary USD bake.